

Power Semiconductors in the Present Day: Evolution, Commercial Capabilities, and Emerging Research Devices

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Abstract—Electronics has developed continuously from the industrial era to the present, and power electronics has become one of its most important branches because it enables the conversion, control, and efficient use of electrical energy. Power semiconductor devices are now essential in mobile phones, computers, electric vehicles, renewable-energy systems, electrical distribution and transmission networks, and emerging transportation technologies. This paper rewrites and organizes a review of the evolution of power semiconductor devices during the twenty-first century, with information reviewed as of 05/08/2024. It summarizes the historical development of the field, compares representative commercial devices according to voltage, current, and switching-frequency capability, and discusses recent research trends based on wide-bandgap and ultra-wide-bandgap materials such as silicon carbide, gallium nitride, gallium oxide, and diamond. The results show that although commercial silicon and silicon-carbide devices already cover a wide range of industrial applications, current research points toward higher power density, higher efficiency, and faster switching in future power-electronic systems.

Index Terms—Power electronics, power semiconductors, MOSFET, IGBT, SCR, wide-bandgap materials, semiconductor history.

I. INTRODUCTION

Power electronics is the branch of electronics dedicated to the conversion and control of electrical energy. Its applications include particle accelerators, amplifiers, household appliances, renewable-energy systems, electric motors, rail systems, aircraft, electric vehicles, and power grids. In all these systems, semiconductor devices such as diodes, transistors, thyristors, MOSFETs, IGBTs, and other controlled switches are fundamental because they determine the operating voltage, current, switching frequency, efficiency, and reliability of the converter.

The development of power electronics has been driven by the need to transform primary electrical energy into useful forms with better efficiency, quality, and controllability. Before the year 2000, progress was strongly associated with the study, patenting, manufacture, and industrial adoption of semiconductor devices. During the current century, this development accelerated because the semiconductor industry pursued smaller, denser, and more capable devices. The effect is visible in the transition from traditional silicon-based components to

silicon carbide (SiC), gallium nitride (GaN), gallium oxide (β -Ga₂O₃), diamond, and other advanced materials.

This paper presents an IEEE-style English version of the original report. Because semiconductor data changes rapidly, the review date is stated as 05/08/2024. It first states the assigned problem, then summarizes the evolution of power semiconductors since 2000, compares representative commercial devices, and finally reviews selected devices currently reported in recent research literature.

II. PROBLEM STATEMENT

The first objective is to prepare a chronological representation, in either two or three dimensions, of the development of power electronics during the current century. The analysis should use information from commercial semiconductor databases, presentations by technology companies, and scientific databases covering approximately the period from 2000 to 2024.

The second objective is to evaluate the current capability of semiconductor devices in terms of voltage, current, and frequency. Two three-dimensional comparisons are required: one for commercially available semiconductor devices and another for devices that are still under research or development. The research-device comparison should be based on recent scientific sources, preferably from 2020 to 2024.

III. EVOLUTION OF POWER SEMICONDUCTORS

The evolution of power semiconductors is closely linked to the evolution of power-electronic applications. As societies became more industrialized, electrical systems required better energy conversion, better efficiency, and higher power quality. This need encouraged the development of semiconductor devices capable of switching and controlling increasingly demanding levels of voltage and current.

In the twenty-first century, the trend toward miniaturization and integration has continued. Moore's law and the concept of "More than Moore" influenced the semiconductor industry by encouraging not only smaller devices, but also devices with improved power density, thermal behavior, robustness, and system-level integration. Today, chips are manufactured at nanometer scales, while power devices are also being

improved through better structures, packaging, thermal management, and new materials.



Fig. 1. Chronological evolution of power semiconductor devices and related power-electronics milestones from 2000 to 2024

Figure 1 summarizes the chronological evolution of power semiconductor devices and related milestones from 2000 to 2024; it was prepared using the historical, commercial, and scientific sources in [1]–[16]. Among the most important advances is the adoption of wide-bandgap materials. SiC is already commercially mature in many high-voltage and high-temperature applications, while GaN is widely studied and used in high-frequency and compact converters. Ultra-wide-bandgap materials such as diamond and β -Ga₂O₃ are promising because they could support higher electric fields, faster switching, and improved power density.

TABLE I
COMMERCIAL POWER MOSFETs

Model	V_{DSS} (V)	I_D (A)	Freq. (MHz)
Nexperia PSMNR90-40YSNX	40	320	≈ 4.7
Vishay Siliconix SQJQ142E	40	345	≈ 8
Infineon IPW60R045P7	650	61	≈ 7.6
Microchip APT10035LFLG	800	28	≈ 4.3
Toshiba TW015N120C	1200	100	≈ 2.4
Microchip MSC035SMA170S	1700	59	≈ 10

IV. CURRENT CAPABILITIES OF COMMERCIAL DEVICES

The following subsections summarize representative commercial power semiconductor devices. The selected parameters are blocking voltage, current capability, and an estimated switching frequency. Most of the commercial device information presented in this section was obtained from component-distributor and manufacturer sources, primarily Mouser Electronics, Infineon Technologies, Microchip Technology, Toshiba, and Littelfuse [17]–[21]. In many commercial datasheets the switching frequency is not provided directly; therefore, in the original work, approximate values were inferred from the available switching-time parameters.

A. Commercial Power MOSFETs

Table I lists representative commercial power MOSFETs, and Fig. 2 presents the same values in graphical form. The comparison uses drain-source breakdown voltage (V_{DSS}), drain current (I_D), and approximate switching frequency.

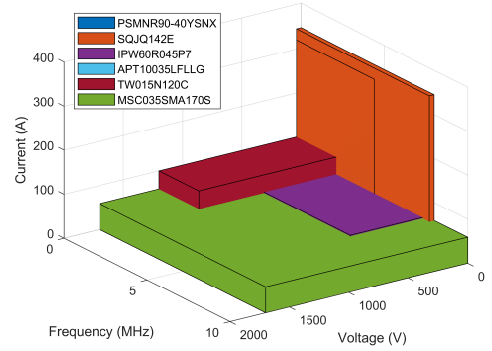


Fig. 2. Three-dimensional comparison of commercial power MOSFET drain-source voltage, drain current, and estimated switching frequency

B. Commercial IGBTs

Table II presents representative commercial IGBTs, while Fig. 3 provides the corresponding graphical comparison. The parameters are collector-emitter voltage (V_{CES}), collector current (I_C), and approximate switching frequency.

C. Commercial SCR Thyristors

Table III summarizes representative silicon-controlled rectifiers (SCRs), and Fig. 4 illustrates their voltage, current, and frequency ranges. The comparison uses repetitive peak reverse voltage (V_{RRM}), RMS direct current, and approximate operating frequency.

TABLE II
COMMERCIAL POWER IGBTs

Model	V_{CES} (V)	I_C (A)	Freq. (MHz)
Diotec DIW075M065	650	90	≈ 2
IXYS IXYX40N450HV	4500	95	≈ 0.5
IXYS IXBX50N360HV	3600	125	≈ 0.3
IXYS IXGK400N30A3	300	400	≈ 1.5
IXYS IXYN140N120A4	1200	380	≈ 0.7
IXYS IXGK75N250	2500	170	≈ 1

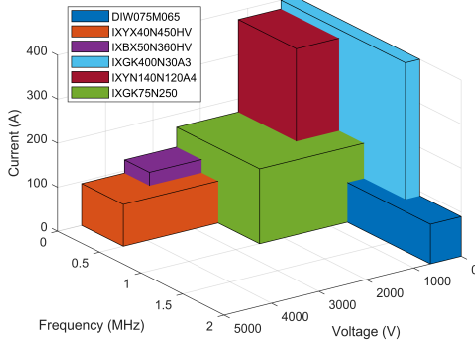


Fig. 3. Three-dimensional comparison of commercial IGBT collector-emitter voltage, collector current, and estimated switching frequency

TABLE III
COMMERCIAL POWER SCRs

Model	V_{RRM} (V)	I_D (A)	Freq. (kHz)
Vishay VS-ST330C16L1	1600	1230	≈ 9.9
Infineon T880N18TOF	1800	2020	≈ 4
Vishay VS-ST700CL Series	2000	1857	≈ 6.6
Vishay VS-ST330C Series	400	1420	≈ 9.9
Vishay VS-ST780CL Series	600	2700	≈ 6.6
Vishay VS-ST650C22L0	2200	1557	≈ 5

TABLE IV
COMMERCIAL POWER DIODES

Model	V_{RRM} (V)	I_F (A)	Freq. (MHz)
Microchip APT30DQ120KG	1200	15	≈ 2.7
Onsemi RURD4120S9A-F085	1200	4	≈ 11
Vishay RGP30M-E3/73	1000	3	≈ 2
ROHM RFN1LAM7STFTR	700	0.8	≈ 12.5
Infineon IDW80C65D1XKSA1	650	80	≈ 6
Infineon IDW75E60	600	120	≈ 5.5

D. Commercial Power Diodes

Table IV shows representative commercial power diodes, and Fig. 5 plots their principal electrical parameters. The parameters are repetitive peak reverse voltage (V_{RRM}), average forward current (I_F), and approximate frequency.

E. Overall Commercial Device Comparison

After the individual device families are reviewed in Tables I–IV and Figs. 2–5, Fig. 6 presents an overall comparison of state-of-the-art commercial silicon power devices based on the latest-generation device comparison reported in [22]. The figure is used as a general benchmark for the voltage and current limits achieved by single-package devices, rather than as a comparison associated with one specific device family.

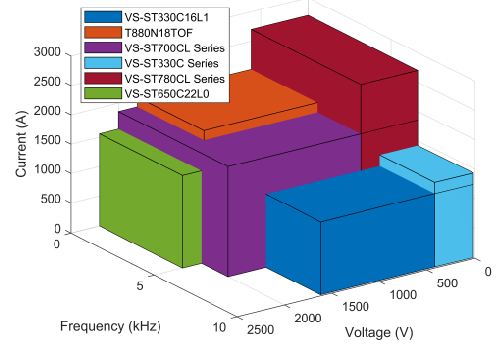


Fig. 4. Three-dimensional comparison of commercial SCR repetitive peak reverse voltage, RMS current, and estimated operating frequency

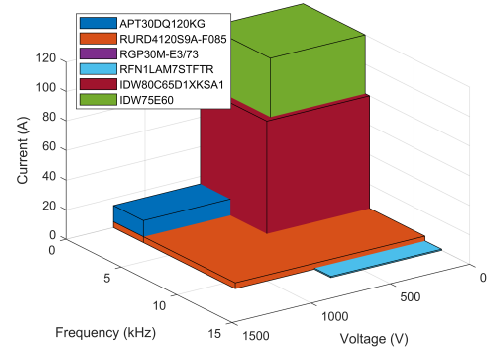


Fig. 5. Three-dimensional comparison of commercial power diode repetitive peak reverse voltage, average forward current, and estimated operating frequency

V. SEMICONDUCTOR DEVICES UNDER RESEARCH

Recent research in power semiconductors is increasingly focused on wide-bandgap and ultra-wide-bandgap materials. Compared with conventional silicon, materials such as SiC, GaN, diamond, and β -Ga₂O₃ exhibit higher critical electric fields, wider bandgaps, improved high-temperature capability, and the potential for higher switching-frequency operation. These properties are relevant for compact high-efficiency converters, high-voltage power modules, radio-frequency power stages, renewable-energy interfaces, electric-vehicle traction systems, and future electrical infrastructure.

Figure 7 summarizes the relative position of emerging power-electronic materials and devices, using the wide- and ultra-wide-bandgap device discussion in [22] and the diamond Schottky diode information reported in [23]; where current information was not directly available, the approximation was supported by the diamond diode characterization study in [24]. The figure is kept before the research-device table to preserve the logical sequence of the original document: first, a qualitative comparison of materials is presented; then, representative experimental or recently reported devices are listed numerically.

Table V summarizes representative power semiconductor devices reported in recent research literature, and Fig. 8 com-

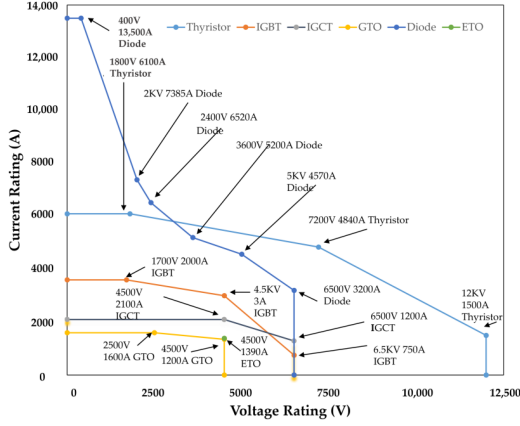


Fig. 6. Overall voltage-current rating envelope of state-of-the-art commercial silicon power devices in single-package implementations

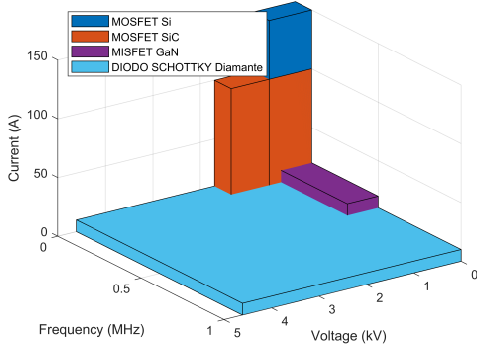


Fig. 7. Comparative capability map of emerging semiconductor materials for power-electronic device technologies

TABLE V
RECENT RESEARCH POWER SEMICONDUCTOR DEVICES

Device	V_{max} (kV)	I_{max} (A)	Freq. (GHz)
β -Ga ₂ O ₃ Schottky diode [25], [26]	3.000	15	≈ 1.4
GaN transistor [27]	0.650	300	≈ 0.1
β -Ga ₂ O ₃ MOSFET [28]	0.192	100	≈ 48
SiC MOSFET [29]	0.970	150	≈ 0.5
Rapid 2 diode [22]	0.650	200	≈ 2
Diamond HEMT [30]	0.121	500	≈ 17
T-gate shallow ion-implanted channel device [30]	0.193	165	≈ 35

compares these research-stage devices graphically. The individual device entries are cited directly in the table to preserve traceability to the original sources. The parameters are organized according to maximum voltage, current capability, and frequency in order to maintain consistency with the commercial-device comparison. Because several devices are still at the prototype or laboratory-demonstration stage, some values are approximate or inferred from the reported material and device characteristics.

The comparison indicates that commercial devices provide mature and reliable solutions for present industrial applications, whereas research devices demonstrate the direction of future performance improvements. In particular, the reported

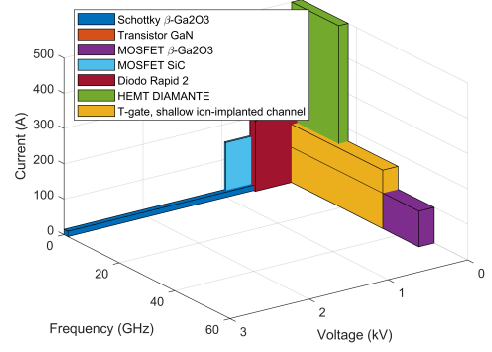


Fig. 8. Three-dimensional comparison of recently reported research-stage power semiconductor devices by voltage, current, and frequency capability

values suggest opportunities for higher switching frequency, increased power density, and operation under more demanding electrical and thermal conditions. Nevertheless, these devices should not be interpreted as direct commercial substitutes because many of them are prototypes, laboratory-scale demonstrations, or structures optimized for a specific performance metric rather than for complete converter-level reliability.

VI. CONCLUSIONS

The review confirms the importance of power electronics in the past, present, and future of electrical engineering. Power semiconductors have enabled the development of compact, efficient, and controllable energy-conversion systems. Their evolution has supported many modern applications, including renewable-energy conversion, electric transportation, industrial drives, consumer electronics, and power transmission and distribution.

The comparison between historical and current devices shows a large technological gap. Devices have become smaller, more efficient, and capable of operating at higher voltage, current, and frequency levels. This progress has been possible because of advances in semiconductor materials, manufacturing processes, packaging, thermal design, and system integration.

The future of power semiconductors is strongly connected to new materials. SiC and GaN are already important in modern power electronics, while diamond, graphene, AlN, BN, ZnO, β -Ga₂O₃, GaAs, and AlGaIn remain active areas of research [31], [32]; additional material properties and comparisons are reviewed in [33]. These materials may help overcome the physical limits associated with device miniaturization and may enable converters with higher efficiency and greater power density.

No single semiconductor device can be considered universally superior. MOSFETs, IGBTs, SCRs, diodes, and emerging wide-bandgap devices each have advantages and disadvantages. The correct device depends on the application requirements, including voltage, current, switching frequency, thermal constraints, cost, reliability, and control strategy.

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